Constraint inequalities

1

a $x \geq 0, y \geq 0$ **b** $55x + 65y \leq 800$ **c** $2x + 7y \leq 200$

2

a $0 \leq 5x + 8y \leq 40$ **b** $0 \leq 10x + 4y \leq 40$

3

a $x \geq 0, y \geq 0$

b $x + y \leq 200$

c The number of fluorescent light bulbs produced must be at least half as many as the number of incandescent light bulbs.

4

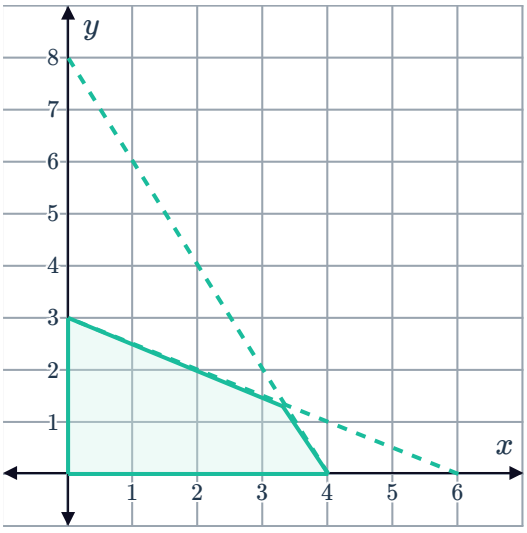
a $2x + 4y \leq 12$

b $3x + 1.5y \leq 12$

c No, x and y cannot have negative values as they are measuring number of pallets made in a day, of which the smallest value would be 0.

d $x \geq 0, \ \ y \geq 0$

e



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

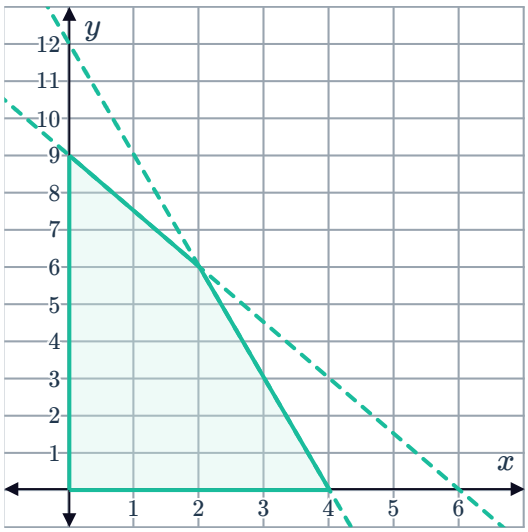
5

a $3x + 2y \leq 18$

b $4.5x + 1.5y \leq 18$

c $x \geq 0,\ \ y \geq 0$

d



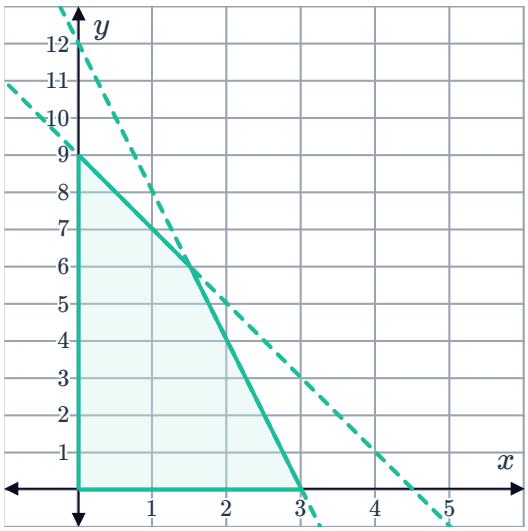
6

a $4x + 2y \leq 18$

b $6x + 1.5y \leq 18$

c $x \geq 0,\ \ y \geq 0$

d

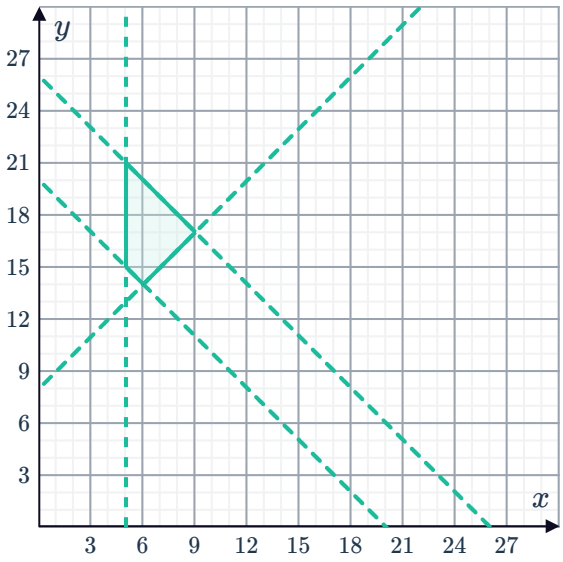


© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7

a $x \geq 5,\ y \geq x + 8,\ x + y \leq 26,\ x + y \geq 20$

b



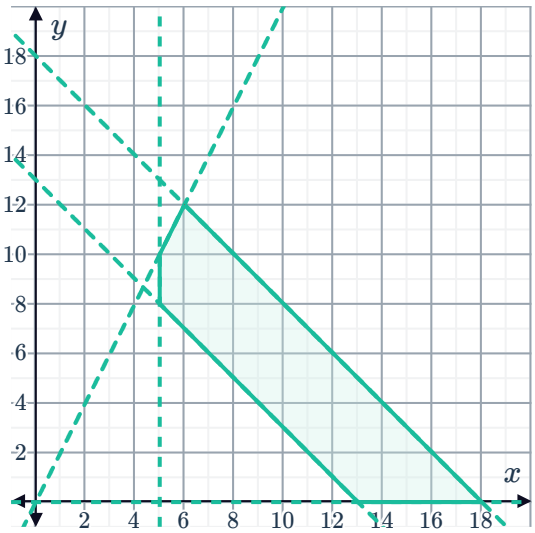
c $(5, 15), (5, 21), (6, 14), (9, 17)$

d 9 sessions

8

a $x \geq 5,\ y \geq 0,\ y \leq 2x,\ x + y \leq 18,\ x + y \geq 13$

b



c $(5, 8), (5, 10), (6, 12), (18, 0), (13, 0)$

d 12 lessons

Objective functions

9

a $R = 5x + 12y$ **b** $R = 3x + 3.5y$ **c** $C = 25x + 30y$

10

a $x \geq 0,\ y \geq 0$

b $12x + 10y \leq 40$

c $11x + 9y \leq 35$

d $R = 1400x + 900y$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

© ClipSAT: By Mr. Mohamed Abdallah +966597688647